

The 300-Project Curriculum, Rebuilt

Same count. Same concepts. Every project now produces something you can show.

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How this version differs

The original 300 taught you computer science. This 300 teaches you the same concepts and leaves you holding a portfolio. Nothing was cut for being too hard. Things were cut for producing nothing a hiring engineer or an admissions panel would read.

What was removed	The concept it taught	What now teaches that concept
Contact book, todo list, diary, notes app, password manager (5 projects)	Dictionaries, CRUD, JSON persistence, file organisation	TLE catalogue manager, experiment configuration system, ephemeris reader, telemetry log analyser. Same skills, and you import all four again later
Hangman, quiz, rock paper scissors, guessing game, calculator (5 projects)	Loops, conditionals, state, input validation, randomness	Command-line framework, data downloader with retries, Monte Carlo simulator, unit and time libraries with test suites
Linked list, stack, queue, hash table, binary tree from scratch (5 projects)	Pointers, references, abstract data types	Moved into C and C plus plus, where pointers are real: dynamic array in C, octree allocator, Barnes-Hut tree. Your NeetCode 150 track already covers the algorithm side
Eight web foundation projects, JWT auth, OAuth, React full stack	DOM, HTTP, async, forms, auth, client-server	FastAPI scientific service, Streamlit application, browser 3D orbit viewer, data downloader. Web as infrastructure for science, not as a subject
Ten game development projects	Game loops, input, sprites, UI, audio, saving	3D visualisation with PyVista, GLSL shaders, custom physics engine, procedural terrain. Rendering skills that transfer to scientific visualisation
Telemedicine, mental health screening, nutrition planner, job market predictor and nine other social applications	Full-stack product building, NLP, classification	Phase 9 Earth observation. Same social mission, but computational, and it points at Simera Sense, Dragonfly, Pixxel and the AfSA and ESA course you applied to
Twelve of the fifteen capstones	Systems integration	Three capstones plus a continuous open source and publication track that starts in month three instead of year five

What was added, and why

- Twenty-eight compiled language projects. C, C plus plus, Rust, Fortran, embedded. The original 300 contained two. ICTP MHPC requires a compiled language and Linux. Flight software at every company on your target list is C or C plus plus. This was the largest gap in the document.
- Twenty-two high performance computing projects, up from two. OpenMP, MPI, CUDA, profiling, scaling studies, and a written performance report as the final deliverable.
- Eighteen guidance, navigation and control projects, up from roughly four. This is your strongest single pitch to Isar, OHB, SSTL, Open Cosmos and Astroscale.
- Ten verification and validation projects across Phases 2 and 5. Manufactured solutions, convergence order, energy conservation, Horizons comparison, SGP4 test vectors. A simulator without a validation plot is an animation.
- A research practice phase that ends in written documents, and an open source track that runs continuously rather than sitting at project 294.

The five rules that convert any project here into a CV line

Apply these and a mediocre project becomes evidence. Skip them and a brilliant project stays invisible. This is the most important page in the document.

- Real data, never synthetic. Kepler light curves, Celestrak TLEs, JPL Horizons ephemerides, Sentinel-2 scenes, Gaia catalogues, NOAA feeds.
- Validated against a reference. Ship an error plot against ground truth or a published value, and state the tolerance you achieved.
- Tested and continuously integrated. pytest or ctest, a GitHub Actions workflow, a visible green badge.
- Documented as if a stranger must run it. Physics, method, assumptions, limitations, and one command that reproduces every figure.
- Published where a recruiter can reach it. Streamlit, Hugging Face Spaces, PyPI, crates.io or Zenodo with a DOI. A private repository is not a portfolio.

Corollary: you do not need all 300 to be employable. You need about twenty built to these five rules. The other 280 are how you get good enough to build those twenty well.

The thirteen phases at a glance

Phase	Focus	Project numbers	Count
0	Programming Foundations, Rebuilt	1 to 18	18
1	Compiled Languages and Systems Programming	19 to 46	28
2	Mathematical and Numerical Foundations	47 to 78	32
3	Scientific Software Engineering and Data	79 to 102	24
4	Classical Mechanics and Dynamics	103 to 124	22
5	Orbital Mechanics and Astrodynamics	125 to 154	30
6	Guidance, Navigation, Control and Flight Software	155 to 172	18
7	High Performance Computing	173 to 194	22
8	Machine Learning for Science	195 to 226	32
9	Earth Observation and Remote Sensing	227 to 250	24
10	Computational Physics and Advanced Methods	251 to 276	26
11	Astronomy Data and Research Practice	277 to 292	16
12	Open Source, Publication and Capstones	293 to 300	8

Total: 300 projects. Phases 0 to 3 are the toolkit, roughly a year. Phases 4 to 7 are the core of the career. Phases 8 to 11 are where you produce results. Phase 12 runs continuously from month three.

Suggested ordering

Phase numbers are not a strict sequence. Run three tracks in parallel so that no single stretch is all foundations and no application deadline catches you empty-handed.

Track one, always running: Phase 0 then 1 then 7

Toolkit, then compiled languages, then HPC. This is the track that unlocks ICTP MHPC and flight software roles, and it is the one you are furthest behind on. Start Phase 1 within three months, not within three years.

Track two, always running: Phase 2 then 4 then 5 then 6

Numerical methods into mechanics into astrodynamics into GNC. This is the track that produces your headline projects. Your current orbit predictor already sits in Phase 5.

Track three, always running: Phase 3 and Phase 12

Software engineering discipline and open source contribution, applied to whatever you happen to be building. Never a dedicated block, always a background habit. Your first merged pull request should land before the end of 2026.

Later: Phases 8, 9, 10, 11

Machine learning, Earth observation, advanced physics and research practice. Phase 9 moves earlier if the AfSA and ESA training course selection comes through, because it becomes immediately relevant. Phase 10 moves earlier if you decide astrophysics rather than space engineering is the destination.

The 300 projects

Each entry states what it teaches and what it is worth on your CV. Where a project is marked kept, it survives from your original curriculum, sometimes with a sharper finished state.

Phase 0: Programming Foundations, Rebuilt (18 projects)

Replaces the contact books and hangman games. Every project here outputs a module you will import in Phases 4 to 11. You are not learning Python twice, you are building your own scientific standard library.

No.	Project	What it teaches	What it is worth
1	Scientific units and dimensions library	Classes, operator overloading, dunder methods, invariants	Imported by every later project. First entry in your own toolkit package
2	Physical constants module with uncertainties	Floats, error propagation, immutable data, parsing CODATA	Uncertainty handling is a scientific skill most programmers never learn
3	Time systems converter: UTC, TAI, GPS, TT, JD, GMST	Modular arithmetic, leap seconds, datetime, edge cases	Hard dependency of every orbital project. Getting time wrong is the classic beginner failure
4	Coordinate frame converter: ECI, ECEF, geodetic, topocentric	Trigonometry, rotation matrices, function design	Used in tracking, ground stations and remote sensing alike
5	TLE parser and catalogue manager	Fixed-width parsing, checksums, validation, dictionaries	Your propagator, debris analysis and ground station all consume this
6	Ephemeris reader for JPL Horizons output	File I/O, tokenising, defensive parsing, exceptions	Gives you ground truth for every validation plot you will ever make
7	Telemetry log analyser	csv module, statistics, missing data, aggregation	Same skill as the cut CSV project, but on data you will actually own
8	Structured logging and diagnostics module	logging module, levels, handlers, formatters	Every serious codebase has one. Yours will look professional from project 8 onward
9	Experiment configuration system with YAML or TOML	Config parsing, schema validation, defaults, reproducibility	Separating config from code is what makes results reproducible
10	Command-line framework for your own scientific tools	argparse, subcommands, exit codes, help text	Turns scripts into tools. Replaces the cut calculator and converter projects
11	Test suite for your units and time libraries	pytest, fixtures, parametrisation, float comparison	The habit that makes everything after this trustworthy
12	Numerical assertion helpers and tolerance utilities	Machine epsilon, ULP, relative vs absolute error	Teaches why $0.1 + 0.2$ is not 0.3 , in a form you will reuse constantly
13	Bash toolkit for scientific workflows	Shell scripting, pipes, cron, rsync, exit codes	Kept from the original list. Essential for any cluster work later
14	Linux environment built from scratch with dotfiles	PATH, shells, SSH keys, environment modules	ICTP MHPC lists solid Linux knowledge as a requirement. This is that
15	Git fluency drill: branch, rebase, bisect, resolve	Version control beyond commit and push	git bisect finding a real bug in your own history is an interview story
16	Data downloader with caching, retries and rate limits	requests, HTTP status codes, exponential backoff, caching	Replaces the cut API and scraper projects. Feeds Celestrak, Horizons and MAST
17	HDF5 and NetCDF reader and writer	Binary scientific data formats, chunking, metadata	Simulation output is never CSV. Learn the real formats early
18	FITS file reader and header explorer	Astronomy's standard format, astropy.io.fits, WCS basics	Prerequisite for every astronomy project in Phase 11

Phase 1: Compiled Languages and Systems Programming (28 projects)

The single largest gap in the original curriculum. ICTP MHPC requires C, C plus plus or Fortran. Flight software at Isar, OHB, SSTL and Open Cosmos is C or C plus plus. This phase is non-negotiable and should start within three months.

No.	Project	What it teaches	What it is worth
19	Memory model explorer in C	Pointers, stack vs heap, addresses, sizeof	Your first compiled artefact. The mental model everything else rests on
20	Dynamic array (vector) implementation in C	malloc, realloc, free, ownership, growth strategies	Replaces the cut linked list project and teaches far more
21	1D heat equation solver in pure C	Arrays, loops, numerical stencils, output formatting	Your first physics code in a compiled language
22	Multi-file C project with a proper Makefile	Compilation units, headers, linking, build automation	Understanding the build is half of systems programming
23	Debugging drill with gdb and valgrind	Breakpoints, backtraces, memory leaks, invalid reads	A clean valgrind report on your own code is a demonstrable standard
24	Unit testing in C with Unity or Check	Test harnesses in a language without pytest	Very few self-taught engineers test their C. This distinguishes you
25	Fixed-point arithmetic library in C	Integer scaling, overflow, precision without an FPU	Directly relevant to flight software on microcontrollers
26	Vector and quaternion maths library in C	Structs, const correctness, API design, numerical care	Reused by your embedded ADCS work in Phase 6
27	Octree node allocator in C	Recursive structures, pools, memory locality	Foundation of the Barnes-Hut solver later in this phase
28	Binary telemetry frame packer and unpacker	Byte order, struct packing, binary file I/O	How real spacecraft data actually moves
29	CCSDS packet encoder and decoder with CRC	Bit manipulation, masks, checksums, standards compliance	CCSDS is the actual space data standard. Naming it in an interview lands
30	Transition to C plus plus: the heat solver rewritten with RAI	References, constructors, destructors, scope-bound resources	The clearest demonstration of why C plus plus exists
31	Generic numerical containers with templates	Templates, type deduction, compile-time polymorphism	Where C plus plus stops being C with classes
32	Direct-summation N-body in C plus plus	Object design, floating point performance, $O(n^2)$	The baseline you will beat three times over
33	Barnes-Hut octree N-body in C plus plus	Spatial trees, approximation, algorithmic complexity	$O(n \log n)$ on a real physics problem. The classic HPC portfolio piece
34	Linear algebra with Eigen	Expression templates, using a real numerical library well	Every C plus plus scientific codebase uses Eigen or something like it
35	ODE integrator library in C plus plus	Policy-based design, abstraction without overhead	Becomes the engine under your orbital propagator
36	CMake project with tests, install targets and CI	Modern build systems, targets, dependencies	Nobody hires a C plus plus engineer who cannot drive CMake
37	GoogleTest suite for the integrator	Fixtures, death tests, floating point matchers	Test coverage on numerical code is rare and impressive
38	Benchmark harness with Google Benchmark	Microbenchmarking, statistical noise, warmup	Measuring before optimising is the whole discipline
39	Static analysis and sanitiser pass	clang-tidy, AddressSanitizer, UndefinedBehaviorSanitizer	Catching undefined behaviour in your own code is a maturity signal
40	pybind11 bindings for your C plus plus integrator	Foreign function interfaces, build integration, GIL	Exactly what research software engineers are paid to do

No.	Project	What it teaches	What it is worth
41	Rust fundamentals through a vector maths crate	Ownership, borrowing, lifetimes, Result	Isar Aerospace uses Rust. So does an increasing share of the sector
42	Rust TLE parser and SGP4 propagator on crates.io	Modules, traits, error handling, publishing	A published crate is externally verifiable evidence, not a claim
43	Fortran comprehension: read and modify a legacy routine	Column-major arrays, implicit typing, common blocks	Half of operational climate and astrophysics code is Fortran
44	ISO C binding: call Fortran from C and from Python	Interoperability, calling conventions, array layout	The skill that lets you work on codes older than you are
45	Cross-language reproducibility study	IEEE 754, compiler flags, fast-math hazards, ordering	A short written report on why your three implementations disagree
46	Embedded C on a Pico or STM32: blink to sensor read	Registers, interrupts, no heap, real-time constraints	Hardware you can hold. Sets up the detumbling controller in Phase 6

Phase 2: Mathematical and Numerical Foundations (32 projects)

Kept almost entirely from your original Phase 1 tier 2 and Phase 3 tier 1, with the pure data structure exercises removed because your NeetCode 150 track already covers them under your no-AI-code rule. Ten verification and stability projects added, because correctness is what separates a numerical scientist from a programmer.

No.	Project	What it teaches	What it is worth
47	Sieve of Eratosthenes with optimisation study	Algorithms, memory access patterns, complexity	Kept. Becomes a vectorisation exercise later
48	Matrix operations from scratch	Nested loops, linear algebra fundamentals, determinants	Kept. Foundation for everything numerical
49	Sorting algorithm benchmarking suite	Complexity in practice, timing methodology, best and worst case	Reframed from visualiser to benchmark. The measurement is the point
50	Floating point behaviour and catastrophic cancellation lab	IEEE 754, rounding, cancellation, summation order	New. Explains most of the bugs you will hit for the next decade
51	Condition number and numerical stability study	Conditioning, backward error, ill-posed problems	New. The vocabulary of numerical analysis
52	Descriptive statistics from scratch	Mean, median, variance, streaming algorithms	Kept. Write Welford's algorithm, not the naive two-pass version
53	Probability distributions and sampling from scratch	PDFs, CDFs, inverse transform, rejection sampling	New. Underpins Monte Carlo and Bayesian work
54	Monte Carlo probability simulator	Random sampling, law of large numbers, convergence	Kept. Monty Hall and birthday problem as validation cases
55	Complex number library with polar form	Operator overloading, atan2, phase	Kept. Needed for FFT and signal work
56	Polynomial solver with Horner and Newton	Efficient evaluation, root finding, discriminants	Kept
57	Gaussian elimination with partial pivoting	Row reduction, pivoting, singular detection	Kept. Then compare against LAPACK and explain the difference
58	LU, QR and Cholesky decompositions	Factorisation, stability, when each applies	New. The workhorses of applied linear algebra
59	Numerical differentiation with error analysis	Forward, backward, central differences, truncation vs round-off	Kept. The h too small paradox is a genuinely deep lesson
60	Numerical integration: Riemann, trapezoid, Simpson	Quadrature, order of accuracy, error scaling	Kept
61	Richardson extrapolation and convergence order estimation	Empirical order of accuracy, error constants	New. How you prove your solver is second order rather than claiming it
62	Automatic differentiation, forward and reverse mode	Dual numbers, computational graphs, chain rule	New. The mechanism behind PyTorch, built by hand
63	Bisection and fixed-point iteration	Bracketing, guaranteed convergence, rate	Kept
64	Newton-Raphson applied to Kepler's equation	Derivatives, convergence rate, divergence cases	Kept and refocused. Handle e greater than 0.9 where naive Newton fails
65	Euler, RK2 and RK4 solvers with order verification	Time stepping, local vs global error, stability	Kept. Verify the order empirically, do not assert it
66	Adaptive step ODE solver with error control	Embedded RK45, step size control, tolerances	Kept. Essential for eccentric orbits
67	Symplectic integrators and energy conservation	Leapfrog, Yoshida, phase space volume, secular drift	New. Run one million steps and plot the energy. This is the plot
68	Method of manufactured solutions verification harness	Formal code verification, source terms, order tests	New. Almost no self-taught engineer knows this exists

No.	Project	What it teaches	What it is worth
69	Stiff ODE solvers and implicit methods	Stiffness, implicit Euler, BDF, stability regions	New. Atmospheric and chemical models are all stiff
70	Boundary value problems by shooting and relaxation	Two-point BVPs, iteration, sensitivity	Kept
71	Conjugate gradient for sparse systems	Krylov subspaces, preconditioning, convergence	Kept. The standard large-system solver
72	Eigenvalue solvers: power iteration and QR algorithm	Iteration, deflation, convergence	Kept and extended
73	Singular value decomposition and low-rank approximation	SVD, truncation, compression	Moved earlier. Image compression as the demonstration
74	Interpolation: Lagrange, cubic spline, Chebyshev	Interpolation error, Runge phenomenon, node choice	Kept. Used for gappy telemetry later
75	Least squares fitting with normal equations and QR	Regression, design matrices, conditioning	Kept. Fit Kepler's third law to real planetary data
76	Optimisation: gradient descent, Nelder-Mead, BFGS	Line search, quasi-Newton, convergence criteria	Kept and expanded. Trajectory optimisation depends on it
77	Constrained optimisation with Lagrange multipliers and penalties	KKT conditions, feasibility, penalty methods	New. Real engineering problems are always constrained
78	Quaternion algebra and rotation library	Non-commutative algebra, slerp, gimbal lock	New here as maths, applied in Phase 6 as attitude control

Phase 3: Scientific Software Engineering and Data (24 projects)

This replaces the eight web foundation projects and the enterprise architecture projects. Everything here exists to make your science reproducible, installable and reachable by a recruiter. Track C from the earlier analysis lives here.

No.	Project	What it teaches	What it is worth
79	pytest suite with fixtures and parametrisation	Test design, isolation, coverage	The floor for any codebase you want taken seriously
80	Property-based testing with Hypothesis on numerical code	Invariants, shrinking, edge case discovery	Finds bugs your example-based tests never will
81	Continuous integration with GitHub Actions and a matrix build	CI pipelines, caching, multiple Python and OS versions	A green badge on your README is instant credibility
82	Package your toolkit: pyproject.toml, wheel, TestPyPI	Packaging, entry points, dependency specification	Turns scripts into software. First step toward the JOSS paper
83	Sphinx documentation with doctests on ReadTheDocs	API docs, narrative docs, executable examples	Public documentation is the difference between a repo and a project
84	Type hints and mypy across a numerical codebase	Static typing, generics, numpy typing	Rare in scientific Python. Immediately noticeable
85	Versioned release with a changelog and git tags	Semantic versioning, release discipline, deprecation	Shows you think about users, not just code
86	Docker image for a reproducible scientific environment	Layers, multi-stage builds, pinning	Kept from Phase 6. Moved much earlier because everything needs it
87	Conda and mamba environments with lockfiles	Dependency resolution, binary packages, reproducibility	The scientific Python reality, whatever you think of it
88	DVC data versioning for a dataset-driven project	Data lineage, remote storage, pipeline stages	Version control for the gigabytes that git cannot hold
89	Snakemake or Nextflow workflow for a multi-step analysis	DAG execution, resumability, provenance	How real computational pipelines are actually orchestrated
90	One-command regeneration of every figure in a report	Build systems for science, determinism, seeds	The single most persuasive thing in a research application
91	Binder-ready repository	Environment specification, cloud execution	A reviewer clicks once and runs your analysis. Nothing beats that
92	Zenodo archival with a minted DOI	Persistent identifiers, citation, archiving	Your code becomes citable. It also becomes a CV line
93	Zarr and cloud-native array storage	Chunking, compression, object storage, lazy access	How Earth observation and simulation data are stored now
94	Parquet and columnar storage for large catalogues	Columnar formats, predicate pushdown, compression	Handling a million-row Gaia subset without loading it all
95	PostgreSQL schema for a satellite tracking database	Normalisation, foreign keys, indexes, EXPLAIN	Kept from the original list. Genuinely useful here
96	Time-series database for telemetry	Retention, downsampling, continuous aggregates	Kept. Years of solar wind data stored properly
97	Task queue for long-running scientific jobs	Workers, retries, scheduling, idempotency	Kept. Periodic orbit propagation as the use case
98	FastAPI service exposing a scientific model	REST design, validation, async, OpenAPI docs	Merges the two kept Flask projects into one modern deliverable
99	Streamlit interactive scientific application	Reactive UI, caching, layout, deployment	Your primary demo delivery mechanism. Cheap and high impact
100	Publication-quality figure standards in matplotlib	Typography, colour maps, accessibility, vector export	Ugly figures undermine good science. This is a real skill
101	3D scientific visualisation with PyVista or VTK	Meshes, volumes, camera control, rendering	Replaces most of the cut game development work

No.	Project	What it teaches	What it is worth
102	Interactive 3D orbit viewer in the browser	Plotly or three.js, animation frames, camera control	Kept. The most shareable artefact of your space work

Phase 4: Classical Mechanics and Dynamics (22 projects)

Kept almost intact from the original Phase 3, with seven diagnostics and chaos projects added. This phase is where your physics becomes computational rather than theoretical.

No.	Project	What it teaches	What it is worth
103	3D vector visualiser	Vectors, dot and cross products, 3D plotting	Kept
104	Projectile motion with atmospheric drag	Kinematics, drag models, terminal velocity	Kept. Rocket ascent through the atmosphere as the twist
105	Simple and double pendulum	ODEs, numerical integration, chaos	Kept. The double pendulum is your first chaotic system
106	Spring-mass oscillator with damping and driving	Second order ODEs, resonance, phase lag	Kept
107	Two-body gravitational simulation	Newtonian gravity, conservation, RK4	Kept. Earth and Moon with real parameters
108	Direct N-body simulation in Python	Force accumulation, $O(n^2)$, softening	Kept. The baseline for your C plus plus version
109	Barnes-Hut in Python and comparison with your C plus plus build	Tree codes, language performance gap	New. The comparison plot is the deliverable
110	Collisional N-body with softening and adaptive timesteps	Close encounters, timestep criteria, energy error	New. Where naive integrators fall apart
111	Integrator comparison on one system: Euler, RK4, Verlet, leapfrog	Order, stability, symplecticity, cost per accuracy	New. A single figure that proves you understand integrators
112	Energy and momentum conservation diagnostics suite	Invariants, drift detection, diagnostics as tests	New. Attaches to every simulation you write afterwards
113	Three-body problem and the figure-eight orbit	Non-integrability, sensitivity, periodic solutions	Kept
114	Phase space portraits and Poincare sections	Phase space, surfaces of section, quasi-periodicity	New. How chaos is actually diagnosed
115	Lyapunov exponent computation	Divergence rates, tangent dynamics, predictability horizon	New. Quantifies chaos rather than just showing it
116	Zero-velocity curves and the Jacobi constant	Rotating frames, effective potential, forbidden regions	New. The bridge to CR3BP in Phase 5
117	Tidal forces and the Roche limit	Differential gravity, tidal disruption, fluid vs rigid	Kept, two original projects merged
118	Perihelion precession with relativistic correction	Secular effects, perturbation terms, small signals	Kept. Reproducing Mercury's 43 arcseconds is a real validation
119	Binary star system with a synthetic light curve	Reduced mass, mass ratio, eclipses	Kept. Connects directly to Phase 11 transit work
120	Coupled oscillators and normal modes	Eigenvalue problems in physics, mode decomposition	Kept. Links linear algebra to dynamics
121	Rigid body dynamics and the intermediate axis theorem	Inertia tensor, Euler equations, gyroscopic stability	Kept. Directly relevant to spacecraft tumbling
122	Lagrangian mechanics solver	Generalised coordinates, Euler-Lagrange equations	Kept. Solve the double pendulum without Cartesian forces
123	Hamiltonian mechanics and symplectic integration toolkit	Canonical equations, phase space volume, long-term stability	Kept. Ten million year solar system integration as the demonstration
124	Custom 2D physics engine	Verlet integration, collision response, constraints	Kept from Phase 4 of the original. Rendering skills that transfer to simulation

Phase 5: Orbital Mechanics and Astrodynamics (30 projects)

The heart of the curriculum and the reason for everything before it. Eleven validation and mission-design projects added, because a propagator without a Horizons error plot is an animation.

No.	Project	What it teaches	What it is worth
125	Kepler's laws demonstrated numerically	Elliptical orbits, areal velocity, period relation	Kept. Verify all three from simulation output
126	Orbital elements and state vector conversion	Six elements, singular cases, frame conventions	Kept. Handle circular and equatorial degeneracies properly
127	Time and reference frame transformations for astrodynamics	Precession, nutation, polar motion, IAU conventions	New. The unglamorous work that makes everything else correct
128	Two-body orbit propagator	Kepler's equation, mean and true anomaly, propagation	Kept. Your August sprint project
129	Propagator validation against JPL Horizons	Ephemeris comparison, RMS error, error budgeting	New. Turns the propagator from a demo into an engineering artefact
130	SGP4 implementation validated against Vallado test vectors	Standards compliance, mean elements, regression testing	New. Publish the test suite. Other people will use it
131	Perturbation modelling: J2, drag, SRP, third body	Gravitational harmonics, non-conservative forces, layering	New. Decompose the error budget contribution by contribution
132	Ground track generator and repeat-groundtrack design	Earth rotation, ascending node, repeat cycles	New. The design problem every Earth observation mission solves
133	Access and pass prediction with elevation masks	Line of sight, topocentric geometry, visibility windows	New. Feeds directly into your SDR ground station in Phase 11
134	Hohmann transfer calculator	Delta-v, transfer ellipse, phasing	Kept
135	Bi-elliptic transfer and the crossover ratio	Three-burn transfers, when Hohmann loses	Kept
136	Inclination and combined plane change manoeuvres	Velocity vector geometry, optimal altitude for plane change	Kept, two original projects merged
137	Delta-v budget tool for a complete mission	Budgeting, margins, Tsiolkovsky, staging	New. What mission analysts actually produce
138	Lambert's problem solver	Two-point boundary value orbits, time of flight, multi-revolution	Kept. Underpins rendezvous and interplanetary design
139	Porkchop plot generator for real launch windows	C3 energy, arrival velocity, launch period scanning	New. One of the most recognisable figures in mission design
140	Interplanetary transfer with patched conics	Spheres of influence, hyperbolic excess velocity	Kept
141	Gravity assist and the Voyager grand tour	Frame transformation, flyby geometry, energy gain	Kept
142	Escape trajectories and hyperbolic orbits	Energy, eccentricity greater than one, asymptotes	Kept
143	Lagrange points and the JWST halo orbit	Rotating frame equilibria, stability, station-keeping cost	Kept. Compare against the published JWST orbit
144	CR3BP trajectory design with invariant manifolds	Jacobi constant, manifolds, low-energy transfers	Kept. Graduate-level and it shows
145	J2 perturbation and sun-synchronous orbit design	Nodal regression, frozen orbits, mission design	Kept. Verify against a real Earth observation satellite
146	Atmospheric drag and orbital decay prediction	Exponential and NRLMSISE atmospheres, ballistic coefficient	Kept. Validate against real TLE decay history
147	Solar radiation pressure and solar sail trajectory	Photon momentum, area to mass ratio, cannonball model	Kept

No.	Project	What it teaches	What it is worth
148	Station-keeping simulator over one year	Control loops, dead-bands, fuel budgets	Kept. A GEO slot held for a simulated year
149	Clohessy-Wiltshire relative motion and rendezvous	Relative dynamics, approach ellipses, safety ellipses	New. Directly relevant to Astroscale and any servicing company
150	Formation flying and constellation relative geometry	Relative orbital elements, drift, maintenance	New
151	Conjunction assessment and probability of collision	Covariance propagation, encounter geometry, screening	Kept and upgraded. Real probability, not just miss distance
152	Launch trajectory and gravity turn optimisation	Staging, thrust profiles, atmospheric losses	Kept
153	Atmospheric reentry dynamics and heating	Ballistic and lifting reentry, peak deceleration, corridor	Kept
154	Complete orbit determination from tracking data	Batch least squares, residuals, observability	New. Closes the loop between estimation and astrodynamics

Phase 6: Guidance, Navigation, Control and Flight Software (18 projects)

Entirely new as a coherent block. This is your strongest single pitch to Isar, OHB, SSTL, Open Cosmos and Astroscale, because it is the subsystem they all need engineers for and almost no applicant at your stage has built one.

No.	Project	What it teaches	What it is worth
155	Attitude representations: DCM, Euler angles, quaternions	Rotation composition, singularities, conversion	The vocabulary of every GNC conversation
156	Sensor models: magnetometer, sun sensor, gyroscope	Noise, bias, drift, scale factor errors	Realistic sensors are what make a simulation credible
157	Star tracker simulator with catalogue matching	Star catalogues, pattern recognition, lost-in-space problem	Genuinely impressive and rarely attempted
158	TRIAD and QUEST attitude determination	Deterministic attitude from two vectors, Wahba's problem	The classical foundation of attitude determination
159	Kalman filter from scratch	Predict and update, covariance, optimality	Kept from the original. Now in its proper context
160	Extended Kalman filter with sensor fusion	Linearisation, Jacobians, multi-sensor integration	Kept. IMU plus GPS plus star tracker
161	Multiplicative EKF for quaternion attitude estimation	Error-state filtering, quaternion constraint handling	The actual algorithm flying on real spacecraft
162	Filter consistency testing with NEES and NIS	Covariance consistency, tuning, statistical validation	Proves your filter is right rather than merely converging
163	PID attitude controller with tuning study	Proportional, integral, derivative, stability margins	Kept and refocused onto spacecraft
164	Reaction wheel model with saturation and momentum dumping	Actuator limits, momentum management, wheel desaturation	The constraint that makes control hard in practice
165	Magnetorquer model and B-dot detumbling law	Magnetic control, geomagnetic field models, damping	Every CubeSat starts with detumbling
166	Full 6-DOF ADCS closed loop simulation	Sensors to estimator to controller to actuators, end to end	Your headline technical project. One figure explains your whole capability
167	B-dot controller in embedded C on real hardware	Fixed-point maths, no dynamic allocation, interrupts	Flight software discipline demonstrated on hardware you can hold
168	Hardware-in-the-loop test rig	Real-time interfaces, timing, test automation	How flight software is actually validated
169	Real-time scheduling and deterministic control loops	Jitter, priority, worst-case execution time	The distinction between fast code and real-time code
170	CCSDS telemetry and telecommand handling	Packet standards, command validation, sequence counts	Speaking the sector's actual protocol
171	FDIR state machine on a telemetry stream	Fault detection, isolation, recovery, safe mode	Operations teams care about this more than about propagators
172	Fault injection and safe mode testing campaign	Failure modes, watchdogs, graceful degradation	A written test campaign report is a portfolio artefact in itself

Phase 7: High Performance Computing (22 projects)

The entry requirement for ICTP MHPC and the skill that makes you useful to any simulation group. Two projects in the original curriculum become twenty-two here, ending in a written performance report that is itself a deliverable.

No.	Project	What it teaches	What it is worth
173	Profiling with perf and gprof	Hotspots, call graphs, sampling vs instrumentation	Measure before you optimise. This is where that habit starts
174	Roofline analysis of a stencil kernel	Arithmetic intensity, memory bandwidth, compute bounds	The roofline plot is the standard HPC diagnostic figure
175	Cache blocking and loop tiling optimisation	Memory hierarchy, locality, tile size selection	Often a five to ten times speedup with no algorithm change
176	SIMD vectorisation with intrinsics and compiler hints	Vector registers, alignment, auto-vectorisation reports	Understanding why the compiler refused to vectorise your loop
177	OpenMP parallel loops and reductions	Shared memory parallelism, work sharing, scheduling	The gateway to parallel programming
178	Race conditions and false sharing laboratory	Data races, atomics, padding, thread sanitiser	Deliberately break it, then diagnose it. That is the lesson
179	Strong and weak scaling study of your N-body code	Speedup, efficiency, scaling methodology	The exact plots an HPC admissions panel looks for
180	Amdahl and Gustafson analysis written up	Theoretical limits, serial fraction, scaling regimes	Turns your measurements into an argument
181	MPI point-to-point communication	Ranks, blocking vs non-blocking, deadlock	Distributed memory from first principles
182	MPI collectives and reductions	Broadcast, scatter, gather, allreduce	The operations every parallel code is built from
183	MPI domain decomposition for 2D Poisson with halo exchange	Ghost cells, boundary communication, load balance	The canonical MPI project. MHPC is built around this pattern
184	Hybrid MPI and OpenMP implementation	Process and thread interaction, affinity, oversubscription	How modern clusters are actually programmed
185	Parallel I/O with HDF5 or MPI-IO	Collective I/O, file striping, the I/O bottleneck	Often the real limit on large simulations
186	CUDA fundamentals: kernels, threads and blocks	GPU execution model, launch configuration	Kept from the original and given proper depth
187	CUDA memory hierarchy and coalescing	Global, shared and constant memory, access patterns	Where GPU performance is actually won or lost
188	CUDA N-body with shared memory tiling	Tiling, occupancy, synchronisation	The textbook GPU optimisation, done properly
189	Barnes-Hut on GPU	Irregular parallelism, tree traversal on device	Substantially harder than the tiled version and worth the effort
190	GPU versus CPU comparative benchmark report	Fair benchmarking, energy and cost per solution	Honest comparison is rarer than you would expect
191	Task-based parallelism with dask	Task graphs, out-of-core computation, schedulers	The Python-side equivalent, needed for Earth observation work
192	SLURM job scripts on a real cluster	Batch scheduling, resource requests, arrays, dependencies	Get access through a university or a free tier and use it
193	Containerised HPC workload with Apptainer	Reproducible environments on shared systems	How research computing centres actually deploy software
194	Performance engineering report as a written deliverable	Technical writing, figures, honest limitations	The document itself is the portfolio piece. Very few applicants have one

Phase 8: Machine Learning for Science (32 projects)

Generative modelling removed because it is a different career. What remains is machine learning that answers scientific questions, on real archival data, with uncertainty quantified.

No.	Project	What it teaches	What it is worth
195	Linear regression from scratch with gradient descent	Cost functions, learning rates, convergence	Kept. Fit exoplanet mass against radius
196	Logistic regression and decision boundaries	Sigmoid, cross-entropy, thresholds, class imbalance	Kept. Habitable zone classification
197	k-means and clustering validity	Centroids, initialisation, elbow and silhouette	Kept. Cluster the HR diagram
198	Decision trees and random forests	Entropy, information gain, bagging, feature importance	Kept. The workhorse for tabular science data
199	Gradient boosting on tabular catalogues	Boosting, regularisation, early stopping	New. Usually the strongest model on tabular data, and honesty about that matters
200	Principal component analysis and spectral decomposition	Covariance, eigenvectors, explained variance	Kept. Reduce stellar spectra to two dimensions
201	Cross-validation and honest evaluation methodology	Leakage, stratification, nested CV, baselines	New. The single most common failure in amateur ML portfolios
202	Time series forecasting with ARIMA and decomposition	Autocorrelation, stationarity, seasonality	Kept. Solar cycle activity
203	Anomaly detection with isolation forests	Outlier scoring, contamination, thresholds	Kept. Unusual satellite telemetry signatures
204	Statistical hypothesis testing for signal detection	Null hypotheses, p-values, multiple comparisons, false alarm rate	Kept. Is that periodic signal actually real
205	Bayesian inference with MCMC from scratch	Priors, likelihoods, Metropolis-Hastings, convergence	Kept. Then move to emcee for real problems
206	Nested sampling and model comparison	Evidence, Bayes factors, model selection	New. How astronomers decide between one planet and two
207	Gaussian process regression with uncertainty	Kernels, marginal likelihood, posterior predictive	Kept. Correlated noise in light curves
208	Neural network from scratch with backpropagation	Forward pass, chain rule, weight updates	Kept. Never use a framework until you have written this
209	Convolutional neural network in PyTorch	Convolutions, pooling, batch norm, augmentation	Kept. Train on real satellite imagery
210	Recurrent networks and LSTMs for sequences	Hidden state, gating, vanishing gradients	Kept. Orbital element evolution
211	Vision transformer for image classification	Patch embedding, attention maps, transfer learning	Kept. Galaxy morphology from Hubble images
212	Autoencoders and variational autoencoders	Latent representations, reconstruction, KL divergence	Kept. Spectral compression and outlier detection
213	Physics-informed neural networks	PDE residual losses, automatic differentiation, collocation	Kept. Compare against your own finite element solution
214	Neural ordinary differential equations	Continuous-depth models, adjoint method	New. Where machine learning and numerical integration meet
215	Uncertainty quantification in deep models	Ensembles, MC dropout, calibration curves	New. A prediction without an uncertainty is not a scientific result
216	Explainable AI for scientific models	SHAP, attribution, counterfactuals, attention	Kept. Why did the model call that galaxy a merger

No.	Project	What it teaches	What it is worth
217	Exoplanet transit classification on real Kepler data	Light curve features, imbalanced classes, CNN on time series	Kept. Your existing sprint project, done to publication standard
218	Crater detection on planetary surfaces	Object detection, bounding boxes, transfer learning	Kept. Real Mars and Moon imagery
219	Galaxy morphology classification with Galaxy Zoo	Large labelled datasets, augmentation, evaluation metrics	Kept
220	Solar flare prediction from magnetograms	Feature extraction, rare event prediction, skill scores	Kept. Rare event metrics matter more than accuracy here
221	Transient detection in survey alert streams	Image differencing, real-bogus classification, throughput	Kept. Synthetic ZTF-like alerts
222	Narrowband signal search in radio data	Spectrograms, interference rejection, anomaly scoring	Kept. Breakthrough Listen sample data
223	Asteroid taxonomy from reflectance spectra	Spectral features, class boundaries, small datasets	Kept
224	Star cluster membership in six-dimensional phase space	Clustering on position and velocity, contamination	Kept. Gaia data makes this genuinely doable
225	Reinforcement learning: Q-learning and DQN	Value functions, exploration, experience replay	Kept, two projects merged. Lunar Lander as the environment
226	Policy gradients and PPO for continuous control	Policy optimisation, advantage estimation, clipping	Kept, merged. Train an autonomous landing and attitude control policy

Phase 9: Earth Observation and Remote Sensing (24 projects)

The original Phase 7 refocused. Health, education and civic applications were admirable but said nothing about you as a computational scientist. Earth observation is where your geography, your AfSA and ESA training application, and Simera Sense, Dragonfly and Pixxel all converge.

No.	Project	What it teaches	What it is worth
227	Raster fundamentals with rasterio and GDAL	Geotransforms, bands, windowed reads, nodata	The base layer of all remote sensing work
228	Coordinate reference systems and reprojection	Datums, projections, resampling, distortion	Getting CRS wrong silently ruins everything downstream
229	STAC catalogue search and asset access	SpatioTemporal Asset Catalogs, queries, cloud access	How modern satellite archives are actually searched
230	xarray and dask pipeline over Rwanda and Cameroon	Labelled arrays, lazy evaluation, out-of-core processing	The exact stack Earth observation companies run
231	Cloud-optimised GeoTIFF generation and tiling	Internal tiling, overviews, range requests	Makes your outputs usable by anyone else
232	Radiometric calibration: digital number to radiance to reflectance	Sensor calibration, solar geometry, gain and offset	The physics under the pretty picture
233	Atmospheric correction implemented from scratch	Radiative transfer, dark object subtraction, aerosol optical depth	Validate against official L2A products. Very few people implement this
234	Spectral indices and vegetation time series	NDVI, NDWI, NBR, temporal compositing	The bread and butter of applied remote sensing
235	Harmonic and seasonal decomposition of vegetation series	Fourier fitting, phenology, anomaly definition	Separates real change from seasonal cycle
236	Change detection: differencing and change vector analysis	Radiometric normalisation, thresholds, false positives	The core analytical operation of the field
237	Land cover classification with random forests	Training sample design, class confusion, accuracy assessment	With a proper confusion matrix and kappa, not just accuracy
238	Cloud and shadow masking with a U-Net	Semantic segmentation, IoU, class imbalance	Kept from the original. Real Sentinel-2 over Central Africa
239	On-board inference constraint study	Quantisation, int8 inference, latency, memory, power budget	Edge processing on satellites is the live problem at Pixxel and Open Cosmos
240	Push-broom optical payload simulator	MTF, SNR, smear, ground sample distance, integration time	Literally what Simera Sense and Dragonfly build. Almost nobody writes one
241	Detector noise and radiometric performance modelling	Shot noise, read noise, dark current, full well	The physics that decides whether an image is usable
242	Point spread function estimation and image restoration	Deconvolution, regularisation, ringing artefacts	Where signal processing meets optics
243	Pan-sharpening and super-resolution	Spatial and spectral trade-offs, evaluation metrics	A visually striking and technically substantive result
244	SAR fundamentals: amplitude, speckle and multilooking	Radar imaging geometry, speckle statistics, filtering	SAR skills are scarce and in high demand
245	Sentinel-1 coherence change detection	Complex data, coherence, temporal decorrelation	Works through cloud, which matters enormously in Central Africa
246	InSAR displacement mapping	Interferograms, phase unwrapping, deformation	Volcanoes, subsidence and infrastructure monitoring
247	Digital elevation models: hydrology and flow accumulation	DEM processing, sinks, flow direction, watersheds	Kept and made properly quantitative
248	Flood risk mapping for Rwanda	Return periods, inundation modelling, exposure	Kept. Real work with a real audience

No.	Project	What it teaches	What it is worth
249	Drought and crop yield prediction	Multi-source fusion, lagged predictors, validation against yields	Kept. Directly useful and directly employable
250	Deforestation monitoring for the Congo Basin	Time series change detection, alerting, validation	Kept. Your mission statement and your employability finally align

Phase 10: Computational Physics and Advanced Methods (26 projects)

The strongest phase in your original curriculum, kept almost untouched. This is the material that distinguishes a computational scientist from a data scientist, and it is what postgraduate admissions panels read most closely.

No.	Project	What it teaches	What it is worth
251	Finite difference solvers for the wave and diffusion equations	Discretisation, stability, CFL condition	The foundation of everything else in this phase
252	Finite element method with mesh generation	Weak formulation, basis functions, assembly	Kept. Stress in a satellite panel, verified by manufactured solutions
253	Finite volume method and conservation laws	Fluxes, Riemann solvers, shock capturing	Kept. Shock tube as the validation case
254	Computational fluid dynamics: 2D Navier-Stokes	Pressure correction, staggered grids, vorticity	Kept. With a grid convergence study, which most tutorials omit
255	Spectral methods for PDEs	Fourier and Chebyshev bases, exponential convergence	Kept
256	Adaptive mesh refinement	Error indicators, refinement criteria, tree grids	Kept
257	Multigrid solver	Restriction, prolongation, V-cycles, smoothing	Kept. A million-point Poisson solve in seconds
258	Particle-in-cell plasma simulation	Charge deposition, field solve, particle push	Kept. Ion thruster physics, relevant to electric propulsion teams
259	Stochastic differential equations and Euler-Maruyama	Wiener processes, Ito calculus, noise	Kept. Cosmic ray transport
260	Molecular dynamics with Lennard-Jones potentials	Neighbour lists, thermostats, equilibration	New. The other great N-body problem, and it hires people
261	Monte Carlo methods in statistical physics	Metropolis algorithm, Ising model, phase transitions	New. Where statistics and physics meet computationally
262	Sparse matrix formats and operations	CSR, CSC, sparse solvers, fill-in	Kept
263	Tensor calculus and index notation toolkit	Metric tensor, Christoffel symbols, contraction	Kept. Prerequisite for the relativity work
264	Special relativity calculator and visualiser	Lorentz transformations, four-vectors, Minkowski diagrams	Kept. Twin paradox as the demonstration
265	Geodesics in Schwarzschild spacetime	Geodesic equation, effective potential, orbits	Kept. Reproduce Mercury's precession as validation
266	Black hole ray tracer	Null geodesics, photon sphere, gravitational lensing	Kept. The most attention-getting image on your portfolio site
267	Kerr metric and rotating black hole orbits	Frame dragging, ergosphere, ISCO	Kept. The natural extension of the previous two
268	Schrodinger equation solver in one and two dimensions	Eigenstates, time evolution, split-step methods	Kept. Quantum tunnelling visualised
269	Quantum harmonic oscillator and ladder operators	Hermite polynomials, coherent states, energy levels	Kept
270	Hydrogen atom and orbital visualisation	Spherical harmonics, radial wavefunctions, degeneracy	Kept. Every orbital from 1s to 3d
271	Stellar structure and evolution	Hydrostatic equilibrium, nuclear burning, the HR diagram	Kept. The Sun from now to red giant
272	Friedmann cosmology and expansion history	Scale factor, density parameters, dark energy	Kept. Reproduce the Lambda-CDM expansion history
273	Gravitational wave detection with matched filtering	Interferometry, strain, templates, signal-to-noise	Kept. Recover a simulated binary merger

No.	Project	What it teaches	What it is worth
274	Wavelet transforms and time-frequency analysis	Multiresolution, denoising, Haar and Daubechies	Kept. Denoise a gravitational wave signal
275	Symbolic mathematics engine	Expression trees, differentiation rules, simplification	Kept. A step-by-step physics equation solver
276	Information theory toolkit	Shannon entropy, mutual information, KL divergence	Kept. Quantify information content in a data stream

Phase 11: Astronomy Data and Research Practice (16 projects)

Where you stop building tools and start producing results. Every project here ends in a figure, a number with an uncertainty, or a written document. This is the phase that makes an application to AIMS, ICTP or a research group credible.

No.	Project	What it teaches	What it is worth
277	Kepler and TESS light curve analysis with lightkurve	Time series, detrending, transit search, BLS periodogram	Kept. Your exoplanet project, done the way astronomers do it
278	Bayesian transit fit with MCMC and corner plots	Priors, posteriors, parameter degeneracies, convergence	Upgrades classification accuracy into a measurement with uncertainties
279	Lomb-Scargle period search on real variable stars	Uneven sampling, aliasing, false alarm probability	New. Time-domain astronomy is where the field is growing
280	Gaia colour-magnitude diagram from the archive	ADQL queries, parallax and distance, extinction, selection effects	New. A beautiful figure and genuine astronomical inference
281	Exoplanet archive statistical population study	Sample selection, detection bias, occurrence rates	Kept. Bias correction is the hard and interesting part
282	ISS tracker with real pass prediction	TLE propagation, topocentric geometry, visibility	Kept. Predict passes over Kigali and then go outside and check
283	RTL-SDR ground station receiving a real weather satellite	RF front end, Doppler correction, demodulation, APT and LRPT decoding	Twenty-five euros of hardware and the best story in any cold email you send
284	Satellite constellation coverage analysis	Revisit time, coverage gaps, Walker constellations	Kept. Real Starlink or Sentinel data
285	Space debris catalogue analysis and risk ranking	Full catalogue processing, conjunction screening, statistics	Kept. Directly relevant to Astroscale
286	Space weather dashboard from NOAA SWPC feeds	Real-time data ingestion, alerting, Kp index	Kept, two original projects merged
287	Aurora visibility forecasting	Geomagnetic activity, auroral oval models, prediction skill	Kept
288	Pulsar timing residuals	Barycentric correction, glitches, timing noise	Kept. The route into nanohertz gravitational wave science
289	CMB angular power spectrum from Planck maps	Spherical harmonics, multipole expansion, masking	Kept
290	Reproduce a published figure end to end from public data	Reading a paper critically, data provenance, honest comparison	The sentence I reproduced Figure 3 of a published paper carries real weight
291	Monthly technical write-up, published publicly	Scientific communication, figure design, stating limitations	Writing decides who gets supervised and who gets ignored
292	Workshop paper or arXiv preprint	Structure, related work, reproducibility statement, peer feedback	The single strongest item you can hold for AIMS, ICTP and MHPC

Phase 12: Open Source, Publication and Capstones (8 projects)

Projects 1 to 5 here should begin in month three and never finish. The three capstones are what you build in your final year. Fifteen capstones was another phase in disguise. Three is already ambitious.

No.	Project	What it teaches	What it is worth
293	First merged pull request to a scientific package	Contribution guidelines, code review, reading someone else's codebase	Externally verified competence. Read directly by Outreachy, ICTP and research groups
294	Substantive feature contribution to the same package	API design, backwards compatibility, design discussion	A typo fix is noise. A feature is a reference you can name in a cover letter
295	Your own package: tested, typed, documented, on PyPI	The full research software engineering lifecycle	Turns your orbital tools into software rather than scripts
296	Journal of Open Source Software submission	Peer review, scientific writing, reproducibility standards	A real, citable, indexed publication needing no laboratory and no supervisor
297	Fully reproducible research repository with a DOI	Containers, pinned environments, one-command reproduction	The standard modern computational science is moving toward
298	Capstone: complete CubeSat mission simulator	6-DOF dynamics, ADCS, power, thermal, comms, ground segment	Integrates Phases 4, 5, 6, 7 and 11 into one defensible system
299	Capstone: African Earth observation analysis platform	Cloud-native pipeline, models, public interface, documentation	Integrates Phases 3, 8 and 9. Your mission statement made operational
300	Capstone: end-to-end exoplanet discovery pipeline	Ingestion, detrending, search, vetting, Bayesian fit, publication	Integrates Phases 2, 8 and 11 and ends in a preprint

What to do this week

- Rebuild the tracker around these thirteen phases. Keep your existing progress by mapping the projects you have already completed onto their new numbers rather than starting from zero.
- Install a C toolchain and write project 19, the memory model explorer. Not next month. This week. It is the difference between the two career paths currently open to you.
- Add the Horizons validation plot to the orbit predictor you are finishing before you deploy it. That is project 23 in Phase 5 and it is a two-day upgrade to work you have already done.
- Choose the one open source package you will contribute to for the next twelve months. One, not five. Read its contributing guide and its issues tagged good first issue. Suggested: astropy, poliastro and its successors, lightcurve, sunpy, skyfield, python-sgp4, rasterio, xarray.
- Set the two fixed deadlines in your calendar now: Flatiron and Simons on 1 December 2026, Janelia in January 2027.

One honest note to close on. Three hundred projects is a decade of work if you do them properly, and you should not expect to finish. That is fine, because the list is not a checklist, it is a map of a field. What matters is that every path through it now leads somewhere. In the original version you could complete a hundred projects and still not have a single artefact that would open a door at Open Cosmos or ICTP. In this version you cannot complete twenty without having several. Build slowly, validate everything, and publish what you build.